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As the popularity and technology of Unmanned Aerial Vehicles (UAVs) continues to improve, it is essential to start exploring new applications to optimize their performance for real-world applications. These advanced UAVs could be utilized to help support organizations such as the United States Forest Service. This document explores a design solution for a UAV capable of supporting the U.S. Forest Service with reforestation efforts and effective radio communication efforts. The research presented in this document proposes the design and implementation of the Seed Transport and Radio Relay (STAR) UAV to fulfill the two stated mission profiles for the US Forest Service. STAR is a UAV designed to be a compact design fitting within a 4-ft x 3-ft x 2-ft box, weigh less than 20 lbs, and be easily assembled and disassembled for efficient transport and operation. To determine the most optimal design of STAR, several trade studies were conducted to select design features, avionic components, and propulsion systems. From these trade studies, the design of STAR was modeled and refined until an optimal configuration was reached. Risk management strategies were then discussed and implemented to address potential hazards that could occur and develop corrective actions. Once the prototype of STAR was manufactured, several Validation and Verification Tests (VV&T) were conducted to ensure the UAV met the design requirements and will perform as expected during flight. Results from these tests will be presented as well as how STAR performed on flight test day and future improvements that can be made to enhance the performance of STAR.

I. Introduction

Unmanned Aerial Vehicle technology has matured rapidly over the past decade, opening the door to applications that extend well beyond traditional military and commercial use cases. One area that stands to benefit significantly is environmental management, where the ability to access remote terrain, cover large areas efficiently, and operate without risk to a human crew addresses challenges that conventional methods struggle to meet. The US Forest Service could heavily benefit from UAV assistance for both restoration efforts and radio communication, as UAVs are capable of accessing remote areas, covering large areas for seed dispersal, and maintaining communication efforts in disaster affected areas saving time and effort.

This document presents the design, development, and testing of the Seed Transport and Radio Relay

UAV, designated STAR. STAR is a fixed-wing vertical takeoff and landing platform developed by Team Apollo at North Carolina State University to address both mission profiles within a single vehicle. The design prioritizes modularity, compactness, and operational simplicity, with interchangeable payload bays and wing configurations that allow the platform to be adapted to either mission without structural modification. This report covers the full development life cycle of STAR, including design overview, risk management, verification and validation testing, and the results of flight test day.

II. Design Overview

The design solution behind the STAR UAV emphasizes both mission readiness and long term adaptability. Developed for the United States Forest Service, STAR has been engineered to meet its core requirements with the highest level of optimization in performance, safety, and operational reliability. At the same time, the system has been deliberately structured as open source and modular, ensuring that future users can customize hardware and software elements to meet evolving mission needs. STAR is built around complete systems integration, combining custom end to end hardware and software to deliver a seamless user experience that minimizes operational complexity while maximizing mission effectiveness. This approach makes STAR not only a robust solution for current forestry applications but also a future proof platform capable of supporting expanded roles in environmental monitoring, resource management, and advanced autonomous operations.

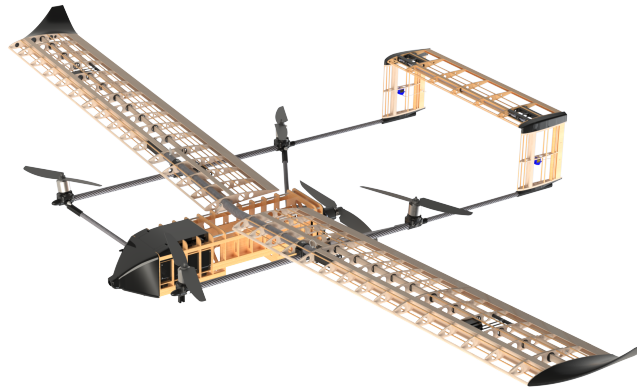


Figure 1: Isometric View CAD Render of Full 10ft Wing Design

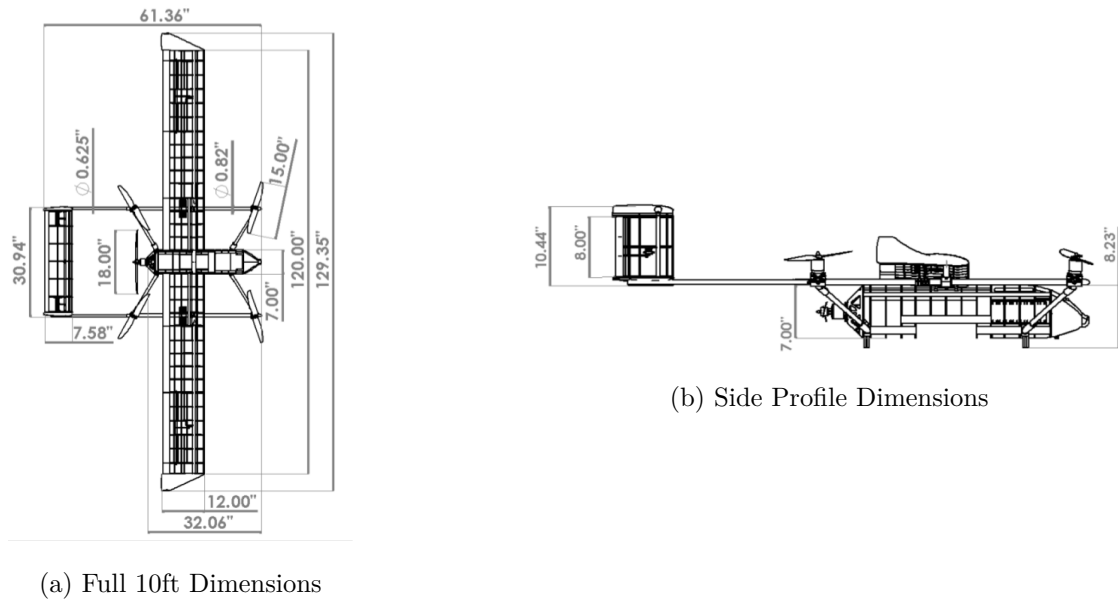


Figure 2: STAR 10ft Configuration Dimensions

A. Key Design Features

1. Flexible Flight Modes

The STAR UAV supports vertical takeoff and landing with smooth transitions to and from forward flight, enabling deployment across both confined and open environments. Wing extensions are fully interchangeable, allowing span adjustments between 10ft and 6ft to optimize endurance, range, and cruise speed for a given mission, and can be removed entirely alongside the tail booms to configure STAR as a dedicated multirotor platform. This hybrid architecture enables takeoff and landing on unimproved terrain, while the ability to hover in VTOL mode supports precision seed dispersal, localized communication relay, and environmental monitoring.

2. Lightweight and Compact Design

STAR employs a lightweight structural philosophy leveraging carbon fiber, machined aluminum, and basswood to achieve an optimal strength-to-weight ratio throughout the airframe. The wing spars utilize a hybrid carbon tube bonded to a basswood I-beam with engineered, permanent 5° dihedral. The fuselage follows a semi-monocoque style construction with basswood bulkheads and longerons, with aluminum rod-rails that support the wing box loading. All structural elements are placed strategically based on loading path and covered with a heat activated adhesive polymer coating. When fully disassembled, the entire system packs within a 4x3x2 foot volume, simplifying logistics for remote deployment without compromising structural integrity.

3. Optimized Propulsion and Power

The VTOL motors deliver a thrust-to-weight ratio as high as 2.5:1, maintained at 1.8 under maximum payload loading in the 10-foot configuration, while the pusher motor is sized for approximately a .7:1 ratio in forward flight. Power is supplied by two custom independent 6S lithium-ion battery packs providing over 700Wh of stored energy and in excess of 300A continuous discharge current, with full operational redundancy in the event of a single pack failure and integrated monitoring and real-time state of charge telemetry delivered to the ground station. The power supply allows all five motors to run in tandem for optimal maneuverability.

4. Flexible Payload and Onboard Safety

STAR incorporates a 2U payload bay centered at the aircraft center of gravity with configurable deployment doors, and a secondary 1U bay directly behind it, both accessible from the top of the fuselage with I2C integration to the central logic unit and electromagnetic shielding. The onboard safety suite includes redundant IMUs, compasses, GPS, and pitot-tube airspeed sensing, and downward LiDAR ranging, with structural sensors on the wing extensions and tail connections enabling failsafe responses including emergency VTOL transition, emergency VTOL rate and attitude assist, and autonomous return-to-home.

5. STARBridge Ground Station

STARBridge is a purpose-built ground station consolidating mission planning, vehicle control, communications, and situational awareness into a single operator interface. Built around a 7-inch touchscreen on an x86 platform, it provides additional controls for flight modes, payloads, and UAV functions alongside live HD video integrated with full telemetry. An integrated HAM radio link keys seamlessly into the onboard repeater, extending operational communication range beyond the UAV itself. Integrated OpenHD architecture allows for encrypted long range communication, and digital HD video for enhanced operational ability and situational awareness.

6. Range and Efficiency

Unlike other designs for multi-rotor vehicles, STAR is able to operate for prolonged periods with long range capabilities. The radio relay mission features an operating radius spanning 10-30 miles while STAR loiters at 400ft AGL, allowing flexible flight patterns for the most efficient cruise conditions. The elevated conditions of STAR allow it to replicate mountain-to-mountain communication, accounting for obstacles and potential interference factors. Combined with the approximated 1.5 hour endurance within the normal flight envelope, STAR provides potential to travel and communicate between cities in favorable flight conditions.

III. Risk Management

Prior to flight operations, a risk assessment was conducted to identify potential failure modes and their associated likelihood and impact on mission success. The following table summarizes the identified risks, categorized by subsystem, with likelihood and impact rated on a scale of 1 to 5, with 1 representing a low likelihood or impact, and 5 representing a certainty or devastating impact.

Risk	Category	Likelihood	Impact
R1: Loss of Communication	Avionics	1	4
R2: Motor/ESC overheating	Propulsion	3	3
R3: Electromagnetic Interference	Avionics	3	4
R4: Wing Connection Failure	Structures	2	5
R5: Control Surface Failure	Stability	2	2
R6: Battery Failure	Avionics	2	5
R7: Individual Motor/ESC Failure	Propulsion	1	4
R8: Electrical Connection Failure	Avionics	2	2

R9: Loss of Control in Transition	Propulsion/Stability	4	3
R10: Pin Failure/Extension Detachment	Structures	1	3
R11: Pusher Prop Strike	Propulsion	1	2

The main concern surrounding the construction and flight testing of STAR are narrowed to failures in propulsion and maintaining connectivity. With redundant batteries and a failsafe hover mode, the points of failure that would result in a total loss of the vehicle would include a motor failure while hovering or the VTOL frame collapsing. While structural considerations are made to ensure no excess strain or cracking will occur on the VTOL frame, individual motor failure can happen instantaneously and sporadically. Therefore, preliminary flight tests will involve a series of examinations to ensure the health of electronic and propulsion components.

IV. Validation and Verification Tests

A. Propulsion

1. Thrust Stand Test

Description:

The thrust stand test provided data on throttle position versus output thrust and current draw, allowing a proof of concept for the propulsion configuration and flight envelope parameters for endurance. The objective of the thrust stand test is to collect data on endurance and thrust figures to calculate the flight envelope of STAR during each mission.

Levels of Success:

- Level 1: Thrust-to-weight ratio calculated is lower than the outlined design requirement, current draw projects Level 1 Endurance.
- Level 2: Thrust-to-weight ratio calculated is equal to the outlined design requirement, current draw projects Level 2 Endurance.
- Level 3: Thrust-to-weight ratio calculated is greater than the outlined design requirement, current draw projects Level 3 Endurance.

Results:

Thrust to weight ratio is above the design requirement, and Level 3 endurance is expected out of the batteries.

B. Aerodynamics

1. Propeller Wash Visualization

Description:

The propeller wash visualization is an experimental test that allows for the analysis of flow direction and strength of the UAV's propellers. The test will validate the effectiveness of the location and size of the propellers as well as verify any downwash effects of the propellers. The objective is to test and verify symmetric, steady flow behavior around propellers when STAR is hovering (VTOL).

Levels of Success:

- Level 1: Recirculation flow around VTOL propellers, turbulent downwash experienced by tuft beneath UAV and on ground, and asymmetric tuft behavior.
- Level 2: No recirculation flow around VTOL propeller, propeller interference between VTOL propellers, and downwash experienced by tuft attached to the UAV and tuft attached to the ground.

- Level 3: Symmetric and steady tuft behavior (minimal downwash) and no propeller interference between VTOL propellers.

Results:

The propeller wash visualization test confirmed steady and symmetric flow generated by the VTOL propulsion system, mainly flow around the propellers without recirculation back into the motors. Also, the strings on the wing were unaffected by the VTOL propellers which confirms that the VTOL propulsion system will not interfere with lift generated by the wings. From this test, the manufacturing process can continue without any changes to the VTOL motors, propellers, and their positioning.

C. CAD & Simulation

1. Center of Gravity Test

Description:

The Center of Gravity Test will verify that STAR's mass distribution will provide stable flight and closely matches the location of the center of gravity modeled in SolidWorks. In order to achieve stable flight it is desired that the center of gravity be in front of or on top of the neutral point. For the test, STAR will be placed on a fixed support and the location at which STAR is balanced represents the center of gravity. This location will be marked using a ruler and a marker and will be compared to STAR's SolidWorks model. Success is measured on accuracy of STAR's center of gravity to the stable model that was run in simulation.

Levels of Success:

- Level 1: CG is within 4 inches of the ideal location.
- Level 2: CG is within 2 inches of the ideal location.
- Level 3: CG is within 0.5 inches of the ideal location.

Results:

After determining the results for the center of gravity test, it can be concluded that STAR meets the requirements for a Level 3 success for both the 10 foot and 6 foot configuration. It meets the Level 3 requirements as both center of gravity locations are within 0.5" of the ideal position.

D. Structures

1. Sandbag Wing Loading Test

Description :

The structural components of STAR must endure loading in VTOL and standard cruise flight modes. The wing and VTOL booms must withstand 3 G-forces and 2 G-forces of loading respectively. These are the maximum assumed conditions that STAR will endure, simulated to examine possible failure or stress points. Physical simulations are conducted to confirm digital results matching STAR's structural capacity of withstanding necessary loading. The objective of this test is to confirm that STAR's manufactured wings and VTOL booms will be able to withstand their respective maximum loading requirements.

Levels of Success :

- Level 1: Wings deform past 2.5 inches, VTOL booms deform past .5 inches. Minimal, repairable damage sustained.
- Level 2: Wings deform exactly 2.5 inches, VTOL booms deform exactly .5 inches. Minimal, repairable damage sustained.
- Level 3: Wings deform less than 2.5 inches, VTOL booms deform less than .5 inches. Nothing breaks.

Results :

The pre-test height of the wing tip was measured to be 49 in. The following parameters were recorded for the loading during the test: actual loading, total = 13,626g (30.04 lbf), effective spar length = 32 $\frac{3}{4}$ " (2.729'),

actual loading, distributed, $w = 10.76 \text{ lbf/ft}$ (.917 lbf/in). The post-test measured height was roughly 45.5 in so the total deflection was calculated to be 3.75 in for a 6ft wing extension (half span). The level of success for this test is Level 2 success because the wing deformed past expected values but no damage was sustained which is a positive outcome.

E. Stability and Control

1. Stability Simulation

Description:

The stability test consists of modeling the wings and tail sections of STAR in AVL. This is performed after STAR is manufactured to ensure model accuracy to the produced result, such as the length of wing and tail that produce zero lift due to a constructed interference such as the fuselage. A fully updated mass file is imported with accurate CG locations, part masses, and moments of inertia. Stability analysis is then performed to show accurate static stability. The success is measured by quantifying how realistic and appropriate stability derivatives and model accuracy are in the simulations

Levels of Success

Level 1: AVL shows that STAR is not stable.

Level 2: AVL shows that STAR is mostly statically stable.

Level 3: AVL shows that STAR is completely statically stable.

Results In both 6 foot and 10 foot wing configurations STAR is statically stable. Tests produced conditions displaying $C_{M_{C_L=0}} > 0$, $C_M < 0$, $C_{l_\beta} < 0$, and $C_{n_\beta} > 0$. Similarly, the sideslip stability derivative $C_{L_\beta} < 0$ and the coefficient of lift with respect to angle of attack conditions $C_{L_\alpha} > 0$ were met. These test results show that STAR achieved the level 3 success criteria for all wing configurations.

F. Avionics

1. Iron Bird Test

Description:

Iron Bird Testing will verify that STAR's electronics and avionics are robust enough for the demands of its varied mission parameters. This method of testing will simulate an extreme flight environment that includes but is not limited to mechanical stresses, vibrations, thermal stress, electromagnetic interference, and electrical strain. In the event that any of STAR's electronics, avionics, or associated systems fail testing; modifications will be made to eliminate points of failure. The objective is to ensure that STAR's avionics and electronics are robust enough to endure rigors of extreme flight conditions.

Levels of Success:

– Level 1: STAR experiences communication and electrical faults to non-critical modules.

– Level 2: STAR experiences minor communication faults to one or more module, but overall systems are functional.

– Level 3: All STAR modules and electronics remain functional throughout testing.

Results:

STAR achieved a Level 3 result on the Iron Bird Test. After mounting into the testing rig, STAR was placed in stabilize mode. A simulated takeoff was then conducted, with STAR remaining in 'hover' for 15 minutes. Temperatures were taken of the motors, ESCS, batteries, and various spots in the wiring to check for thermal stress. No temperature readings were collected over 95 degrees Fahrenheit. While in stabilize mode, STAR was manually disturbed to ensure proper corrective stabilization behavior. Flight controller properly stabilized aircraft, but PIDs led to some oscillatory behavior that showed decaying behavior. STAR successfully simulated both a manual and automated transition from multirotor to fixed wing flight and back. During the test, STAR maintained all system functions, with no faults detected.

V. Flight Test Results

A. First Flight Attempt - Full 10ft Configuration

Preparations for STAR's debut flight were significantly complicated in the twelve hours preceding the event by a complete failure of the primary flight controller. A replacement unit was procured in time for the flight, however the differences in wiring configuration and sensor compatibility required a full avionics rework immediately prior to flight with no opportunity for adequate testing. Among the more consequential changes was the substitution of the encrypted OpenHD video and telemetry system with a SiK clone telemetry radio pair. For its debut flight, STAR was configured for the radio relay mission profile, equipped with the full 10-foot wing configuration and the onboard repeater module.

Immediately prior to takeoff, communication instability was observed between the ground station and the aircraft. At the time, this was not treated as a flight-critical issue, as the intent was to fly STAR in a fully manual capacity without relying on any autonomous mission features from the ground station. Upon takeoff, STAR successfully lifted off in VTOL mode to approximately 10 feet AGL. The aircraft then experienced an uncontrolled pitch moment and descended uncontrollably into the flight field, as shown in Figure 3. Post-flight log analysis revealed that the flight controller was repeatedly receiving MAVLink commands to transition from the intended GPS-stabilized flight mode into a Return-to-Launch condition. As the replacement flight controller had not been fully configured to handle RTL behavior, it responded unpredictably, resulting in the crash and vehicle damage.



Figure 3: First Flight Attempt - 10ft Configuration

B. Second Flight Attempt - Hover Configuration

Immediately following the first flight attempt, a damage assessment was conducted revealing that STAR remained capable of flight in its quadcopter hover configuration with only minor repairs required. Repairs were completed and STAR was prepared for an immediate second attempt given the time constraints of the flight test event.

On the second flight attempt, STAR exhibited stable and well-controlled multirotor flight performance throughout. The vehicle lifted off and ascended to approximately 15 feet AGL, performed a series of flight maneuvers over the field, and landed in a precise and controlled manner as shown in Figure 4. Due to damage sustained during the first attempt, payload system testing and the VTOL-to-fixed-wing transition were not achievable on this flight. Nevertheless, the second flight successfully demonstrated the robustness of the airframe, propulsion system, power architecture, and avionics under real flight conditions.



Figure 4: Second Flight Attempt - Hover Configuration

C. Third Flight Attempt - Full 6ft Configuration

The third flight test was performed with controlled scheduling and extensive safety checks. The third iteration was constructed and programmed to fly with manual controls to account for possible autopilot malfunctions, as seen in the first flight attempt. The frame was assembled and armed in a field, allowing clear LOS manually controlled flight.

The results of the third flight test were a success. The controlled operation of STAR lasted approximately three minutes, and a health inspection of the motors, ESCs and batteries were performed. Further improvements on higher quality VTOL motors and ESCs for proper cooling were noted, since these motors have a consistently high output and thermal measurements indicated that motor overheating would be a potential failure point



Figure 5: Third Flight Attempt - 6ft Configuration

VI. Conclusion

The STAR UAV represents a viable and effective solution to the dual mission requirements required by the United States Forest Service. Through the design and integration of interchangeable payload bays, STAR is capable of fulfilling both a seed dispersal role and a radio relay role within a single platform, eliminating the need for separate purpose-built vehicles for each mission. The modular wing configuration further extends this flexibility, allowing the wingspan to be adjusted based on mission profile and transport constraints. These design decisions, combined with a compact form factor and a fully integrated ground station in STARBridge, result in a system that is operationally practical for deployment in the remote environments the Forest Service routinely operates in.

Testing confirmed that STAR meets or exceeds its core performance requirements across propulsion,

aerodynamics, structures, stability, and avionics. The platform demonstrated stable flight characteristics in both the 6-foot and 10-foot wing configurations, sufficient thrust margin, and robust avionics performance under simulated operational conditions. While further development remains to achieve full mission demonstration, STAR establishes a strong technical foundation as a compact, adaptable, and purpose-built UAV solution for reforestation and communications support in environments where conventional methods fall short.